Floor van Donkelaar

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Institute of Astronomy
University of Cambridge

Main research interests: The formation and evolution of dense stellar systems, particularly stellar clusters and clumps, in high redshift galaxies using (cosmological) simulations.

APPOINTMENTS

Herchel Smith Postdoctoral Research Fellow Oct 2025 -
University of Cambridge, United Kingdom
Institute of Astronomy (IoA) & Kavli Institute for Cosmology, Cambridge (KICC)

PUBLICATION OVERVIEW

As October 2026:
9 papers accepted/submitted with over 115 citations. ADS link for all publications
8 first-author papers accepted with over 92 citations. ADS link for first-author publications
h-index: 6, *g-index*: 8

EDUCATION

PhD in Computational Astrophysics Sep 2021 - Sep 2025
University of Zurich, Switzerland
Thesis: Disc Galaxies and Their Components at High Redshift: Similarities with the Local Universe
Supervisor: Prof. Dr. Lucio Mayer

MSc in Astrophysics Aug 2019 - May 2021
Lund University, Sweden GPA: 3.83/4.00
Thesis: The Fate of Stars Born in Gas-Rich High Redshift Galaxies
Supervisor: Prof. Dr. Oscar Agertz

BSc in Sociotechnical Engineering, Minor in Modern Physics Sep 2016 - Jul 2019
University of Twente (ATLAS), The Netherlands GPA: 3.85/4.00
Thesis: The Star Formation Rate, Metallicity and Thermal Pressure in Galaxies at z=0.4 using MUSE
Supervisor: Dr. Kasper Borello Schmidt from *Leibzin-Institut für Astrophysik Potsdam*

GRANTS & FELLOWSHIPS

Herchel Smith Postdoctoral Research Fellowship ($\sim$ £185K) 2025

EARLY RESEARCH EXPERIENCE

Oxford University
Research Internship Jun 2020 – Nov 2020
Supervisor: Dr. Kearn Grisdale
Analyzing GMCs in the Large Magellanic Cloud with the use of N-body hydrodynamical simulations.

Leibniz-Institut für Astrophysik, Potsdam
Summer Project Jul 2017
Supervisor: Dr. Kasper Borello Schmidt
Generating template spectra of MUSE-Wide emission lines sources.

TEACHING

University of Cambridge, United Kingdom
Presentation Practice Session Tutor Michaelmas Term 2026
Astrophysics Part II Supervisor
Topics in Astrophysics Lent Term 2026

University of Zurich, Switzerland
Teaching Assistant
AST 202: The Universe: Contents, Origin, Evolution and Future Spring 2025, 2024 & 2022
AST 201: Introduction to Astrobiology Fall 2024 & 2021
AST 245: Computational Astrophysics Fall 2024 & 2023
AST 295: Astrobiology proseminar Fall 2023 & 2022
AST 248: The Sun and Planets Spring 2023

SUPERVISING & MENTORING

Master's project
Co-supervisor, Xiaojie Liao, University of Lancashire 2026
"Search for local bar properties in a high-redshift barred simulation"
Co-supervisor, Daniel Swinger, ETH Zurich 2025 - 2026
"Investigating the Birth of Galactic Discs in the Phoebos Simulation"

Bachelor's project
Primary Supervisor, Romy Jager, University of Twente 2026
"Toomre parameter in the galactic disc at $z \sim 7$ within the GigaEris simulation"

AWARDS

Edith Alice Müller Award (Outstanding PhD thesis), Swiss Society for Astrophysics and Astronomy 2026

Pre-doctoral:
University College Twente Third-year Award, University College Twente 2019
More Than A Degree Awards, University of Twente 2018
Communication Award Shell Eco-Marathon, Shell 2018
University College Twente Second-year Award, University College Twente 2018

SERVICE & OUTREACH

Peer Reviewer:
Reviewer for Nature Astronomy 2026 -
Reviewer for Astronomy & Astrophysics (A&A) 2026 -
Reviewer for Monthly Notices of the Royal Astronomical Society (MNRAS) 2024 -

Equality, Diversity & Inclusion:
Co-chair International Women's Day 2026, IoA, University of Cambridge Dec 2025 - May 2026
PhD Representative, Institute of Computational Sciences, University of Zurich May 2022 - Sep 2024
Member faculty council EEMCS, University of Twente Sep 2017 - Aug 2018

Outreach and Public Engagement:
Featured interviewee on dyslexia & astronomy, The Sky at Night, BBC Four Oct 2026
Developed / Co-led a monitor display highlighting research by ECRs, KICC Mar 2026 -
Chief public relations, Green Team Twente Jun 2017 - Sep 2018

SELECTED TALKS

Selected Conference Talks (17 in total, 4 invited)

- (Invited) *'Simulating the First Galaxies: Discs, Bursts, and Clusters'*
  2026 Ringberg Workshop on 'Computational Galaxy Formation', 28 July 2026
- (Invited) *'Clustered Star Formation at High Redshift: A Simulation Perspective'*
  EAS 2026: Stellar clumps and star clusters as building-blocks of galaxy assembly (SS5), 4 Jul 2026
- (Invited) *'What is a Clump? And their correlation to high-redshift discs'*
  Zoom-in Views of galaxy Disks Across Time: Bridging Simulations and Observations, 12 June 2026
- *'Simulating the Early Universe: Shedding Light on Clusters seen by JWST at z>4'*
  Sexten Center for Astrophysics Riccardo Giacconi, 1 Jul 2025
- *'First Results of the Phoebos Simulation: Galaxy Sizes during the Early Universe'*
  SKAO Council meeting (Zurich), 17 Mar 2025
- *'On disc kinematics: The influence of the thin disc on star clusters'*
  Aspen Center for Physics, 18 Mar 2022

Selected Departmental Seminars & Colloquia (12 in total, 3 invited)

- (Invited) *'Simulating the First Galaxies: Stellar Discs and Clumps'*
  Liverpool John Moores University, 21 Oct 2026
- *'The GigaEris Simulation: What makes a high redshift (disc) galaxy different?'*
  Northwestern University, 26 Sep 2024
- *'The GigaEris Simulation: Stellar clusters in MW-sized galaxies at z>4'*
  Flatiron Institute, CCA (New York), 13 Sep 2024
- (Invited) *'Stellar Systems at z > 4: The hybrid formation scenario for the nuclear star cluster'*
  Lund University, 8 May 2023
- *'The Formation of the Nuclear Star Cluster'*
  University of Chicago, 28 Mar 2023

SCIENTIFIC MEMBER

Member of the the JWST **Genesis Initiative**	2026 -
Research Associate at **Emmanuel College**, University of Cambridge	2026 -
Member of the HI Galaxy Science working group of **SKA**	2022 -
Member of **SKACH**, The SKA Switzerland Consortium	2022 - 2025

OTHER ACTIVITIES

Rugby player, Cambridge Rugby Club, competing at a semi-professional level	2025 -